



MANIPAL UNIVERSITY JAIPUR

Department of Mathematics & Statistics

B.Tech. IV Sem.

MAS2001-Statistical Method and Probability theory

Assignment 4

Tests of Statistical Hypothesis: Introduction, Parameter and Statistic, Standard error, Statistical hypotheses, Critical region, Tests of hypotheses and significance, Type I and Type II errors, level of significance.

Section A

1. Define-
 - i). Statistical Hypothesis
 - ii). Null Hypothesis and Alternative Hypothesis
 - iii). One tailed and two tailed test
 - iv). Type I and type II error
 - v). Significance level and critical region
 - vi). Confidence interval and confidence limits.
2. Statistical hypothesis under test is called.....
3. A wrong decision about the null hypothesis leads to types of errors.
4. The probability of type I error is given by.....
5. The level of significance of a test is related to type error and is given by.....
6. Critical region provides us the criterion for of null hypothesis.
7. Which one of the following statement is true for Type II error
 - (a) Rejecting an incorrect Hypothesis
 - (b) Accepting an incorrect Hypothesis
 - (c) Accepting a correct Hypothesis
 - (d) Rejecting a correct Hypothesis
8. An investigator commits a type II error when he / she
 - (a) rejects a null hypothesis when it is true
 - (b) accepts a null hypothesis when it is true
 - (c) rejects a null hypothesis when it is false

(d) accepts a null hypothesis when it is false

9. Type I error in the testing of null hypothesis occurs when the researchers

(a) rejects it, when it is true

(b) accepts it, but it is false

(c) Both (a) and (b)

(d) neither (a) nor (b)

Section B

10 (A) The heights of 10 adult males selected at random from a given locality had a mean of 158 cm and variance 39.0625 cm. State the null and alternative Hypothesis. [Ans: $H_0: \mu=162.5$; $H_1: \mu<162.50$]

10.(B). A local telephone company claims that the average length of a phone call is 8 minutes. In a random sample of 18 phone calls, the sample mean was 7.8 minutes and the standard deviation was 0.5 minutes. Is there enough evidence to support this claim at $\alpha = 0.05$. State the null and alternative Hypothesis.

[Ans: $H_0: \mu=8$; $H_1: \mu \neq 8$]

11. (A) A travel agency's marketing brochure indicates that the standard deviations of hotel room rates for two cities are the same. A random sample of 13 hotel room rates in one city has a standard deviation of \$27.50 and a random sample of 16 hotel room rates in the other city has a standard deviation of \$29.75. State the null and alternative Hypothesis.

11. (B) Consider the following measurements of the heat produced by two mines:

Mine 1	8260	8130	8350	8070	8340	
Mine 2	7950	7890	7900	8140	7920	7840

State the null and alternative Hypothesis.

12. (A) A manufacturer claims that its rechargeable batteries have an average life greater than 1,000 charges. A random sample of 10 batteries has a mean life of 1002 charges and a standard deviation of 14. State the null and alternative Hypothesis

[Ans: $H_0 : \mu = 1000$, $H_1: \mu > 1000$]

12.(B) Certain pesticide is packed into bags by a machine. A random sample of 10 bags is drawn and their contents are found to weight in kgs as follows: 50, 49, 44, 52, 45, 48, 46, 45, 49, 45. State the null and alternative hypothesis that the average packing can be taken as 50kg. [Ans: $H_0: \mu=50\text{kg}$; $H_1: \mu \neq 50\text{kg}$]

Section C

13. Given the frequency function, $f(x, \theta) = \frac{1}{\theta}, 0 \leq x \leq \theta$ and $f(x, \theta) = 0$, otherwise and that you are testing the null hypothesis $H_0: \theta = 1$ against $H_1: \theta = 2$, by means of a single observed value x . What would be the sizes of the Type I and type II errors, if you choose the interval (i) $0.5 \leq x$, (ii) $1 \leq x \leq$

1.5 as the critical regions?

[Ans: (i) $\alpha=0.5$, $\beta = 0.25$, (ii) $\alpha = 0$, $\beta = 0.75$]

14. Let p be the probability that a coin will fall head in a single toss in order to test $H_0: p = \frac{1}{2}$ against $H_1: p = \frac{3}{4}$. The coin is tossed 5 times and H_0 is rejected if more than 3 heads are obtained. Find the probability of type I error.

[Ans: $\alpha=3/16$, $\beta = 47/128$]

15. If $x \geq 1$ is the critical region for testing $H_0: \theta = 2$ against the alternative $\theta = 1$, on the basis of the single observation from the population.

$$f(x, \theta) = \theta \exp(-\theta x), 0 \leq x \leq \infty$$

Obtain the values of type 1 and type II errors.

[Ans: $\alpha=1/e^2$, $\beta = (e - 1)/e$]

16. A random variable X follows the following exponential distribution

$$f(x, \theta) = \frac{1}{\theta} e^{-x/\theta}, 0 \leq x \leq \infty, \theta > 0$$

The null hypothesis $H_0: \theta = 2$ is rejected and alternative hypothesis $H_1: \theta = 4$ is accepted. If the observation selected at random takes the value 6 or more, find the critical region, acceptance region, and the type I and type II error.
